CHAPTER - VII

SEMICONDUCTORS

7.1 INTRODUCTION

Semiconductors are materials which have electrical conductivities lying between those of good conductors and insulators. The resistivity of semiconductors varies from 10^{-5} to 10^{+4} ohm-m as compared to the values ranging from 10^{-8} to 10^{-6} ohm-m for conductors and from 10^{7} to 10^{8} ohm-m for insulators. There are elemental semiconductors such as germanium and silicon which belong to Group IV of the periodic table and have resistivity of about 0.6 and 1.5×10^{3} ohm-m respectively. Besides these, there are certain compound semiconductors such as gallium arsenide (GaAs), indium phosphide (InP), cadmium sulphide (CdS), etc. which are formed from the combinations of the elements of Groups III and V, or Groups II and VI. Another important characteristic of the semiconductors is that they have small band gap. The band gap of semiconductors varies from 0.2 to 2.5 eV which is quite small as compared to that of insulators. The band gap of a typical insulator such as diamond is about 6 eV. This property determines the wavelength of radiation which can be emitted or absorbed by the semiconductor and hence helps to construct devices such as light emitting diodes (LEDs) and lasers. All the semiconductors have negative temperature coefficient of resistance. The band gap energies for the elements of Group IV at 0 K are as follows :

C (diamond)	5.51 eV
Ge	0.75 eV
Si	1.16 eV
Sn (grey)	0.08 eV
Pd	$\cong 0$

At room temperature, diamond behaves as an insulator, Ge and Si as semiconductors and Sn and Pd as conductors.

The importance of semiconductors is further increased due to the fact that the conductivity and the effective band gaps of these materials can be modified by the introduction of impurities which strongly affect their elec-

tronic and optical properties. The process of introduction of impurities in semiconductors in a precisely controlled manner is called *doping*. Depending on the nature of impurities added, the semiconductors are classified as follows :

(a) Pure or intrinsic semiconductors

(b) Impurity or extrinsic semiconductors

The intrinsic semiconductors are pure semiconductors in which no impurity atoms are added, whereas the extrinsic semiconductors are doped semiconductors in which suitable impurity atoms are added to modify the properties. In the present chapter, the effects of impurities and charge carrier concentrations in semiconductors are discussed.

7.2 PURE OR INTRINSIC SEMICONDUCTORS

As stated above, the intrinsic semiconductors such as pure Ge or Si are undoped semiconductors. The electrical conductivity of this type of semicondutor is solely determined by thermally generated carriers. To understand the mechanism of conduction, we consider the bonding between

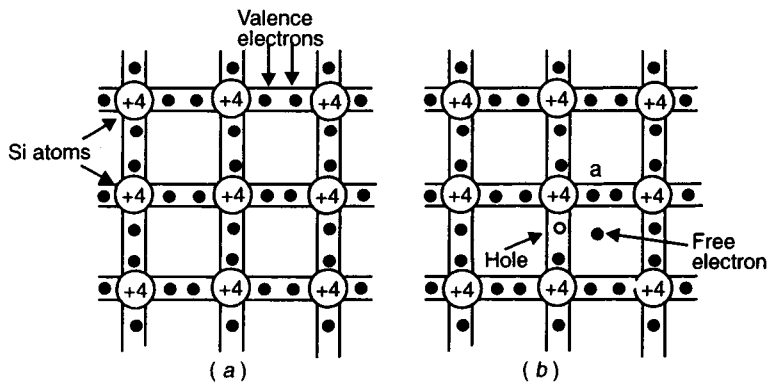


Fig. 7.1. *(a) A two-dimensional representation of Si crystal (b) Silicon crystal containing an electron-hole pair.*

atoms in these semiconductors. Consider, for example, the case of silicon with atomic number 14. Each silicon atom has four valence electrons and can form four covalent bonds with four neighbouring silicon atoms which are directed along the corners of a regular tetrahedron. The silicon crystal, therefore, exhibits a three-dimensional regular network type structure which, for simplicity, is represented by a two-dimensional network as shown in Fig. 7.1a. Apparently, all the valence electrons in a silicon crystal participate in the formation of covalent bonds and no electron is free to cause conduction

particularly at 0 K. Hence pure silicon behaves as an insulator at 0 K. As the temperature increases above 0 K, some of the valence electrons may acquire sufficient thermal energy to break their covalent bonds and become free from the influence of cores of the atoms. These electrons move randomly in the crystal and are referred to as the *conduction electrons.* Each escaped electron leaves behind an empty space called a *hole* which also acts as a current carrier. Thus when a valence electron breaks away from a covalent bond, an electorn-hole pair is generated and two carriers of electricity are produced as shown in Fig. 7.1b. When a valence electron located adjacent to a hole acquires sufficient thermal energy, it may jump into the hole position to reconstruct the broken covalent bond and a hole is created at the initial position of the electron. Thus the motion of an electron may also be regarded as the motion of a hole in the opposite direction. These electrons and holes move in opposite directions under the effect of an external electric field and constitute the current.

The energy band diagram of the intrinsic semiconductor is shown in Fig. 7.2. At 0 K, the valence band is completely filled and the conduction band is completely empty. The semiconductor, therefore, behaves as an insulator. The electrons present in the valence band do not conduct as these are bound to their respective cores. As temperature increases, some of the valence band electrons acquire sufficient thermal energy to jump to the conduction band leaving behind an equal number of holes in the valence band. The electrons in the conduction band and holes in the valence band behave as free carriers and increase the conductivity of the material. The conditions for the movement of electrons are, however, different from the conditions for movement of holes; the electrons move when the conduction band is nearly empty and the holes move when the valence band is nearly full. Thus the electrons move mainly under the influence of the applied field while the holes move under the combined effect of the applied electric field and the ionic field of the lattice. Thus the properties such as effective mass, mobility[1], etc. of a hole are quite different from the corresponding properties of electrons. For example, a hole has larger effective mass and lower mobility than the corresponding values for an electron. The charge of a hole is equal and opposite to that of an electron. These properties have been described in the previous chapter. It is now apparent that, in an intrinsic semiconductor, the number of electrons, n_i, in the conduction band is always equal to the number of holes, p_i, in the valence band, i.e., $n_i = p_i$, and either one of these is called the *intrinsic carrier concentration.*

Besides the generation of free electron-hole pairs, there is another process called *recombination* of carriers in semiconductors. A free electron

1. Mobility of a carrier is the velocity acquired by it in a unit electric field.

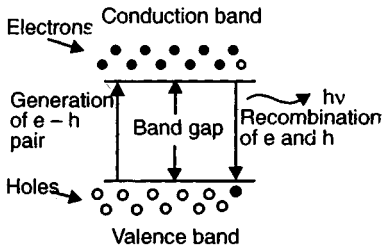


Fig. 7.2. *Energy band diagram of an intrinsic semiconductor showing generation and recombination of an electron and a hole.*

moving randomly in a semiconductor may encounter a hole and combine with it so as to reconstruct the broken covalent bond. Thus the electron-hole pair is destroyed and the free electron is converted into the bound electron. This recombination process is equivalent to an electron jumping from the conduction band to the valence band and occurs with the release of energy equal to the band gap energy in the form of electromagnetic radiation as shown in Fig. 7.2. In an intrinsic semiconductor, the rate of generation of carriers, g, depends on the temperature and nature of the material. The recombination rate, R, on the other hand, depends on the concentration of electrons and holes in a material at that temperature. In equilibrium, the generation rate just equals the recombination rate, i.e.,

$$g = R = C\, n_i p_i$$

where C is a proportionality constant which depends on the nature of the material. Since in an intrinsic semiconductor, $n_i = p_i$, we have

$$g = R = C\, n^2{}_i \qquad (7.1)$$

For a given temperature, the quantity n_i is a constant and depends only on the nature of the semiconductor. It will be realized later that the constant C is related to the densities of states of electrons and holes at the conduction band and valence band edges respectively.

7.3 IMPURITY OR EXTRINSIC SEMICONDUCTORS

The intrinsic semiconductors, as such, are of little importance owing to their very small and fixed conductivity. The introduction of impurity atoms, i.e. doping, is the most efficient and convenient method of increasing and altering the conductivity of an intrinsic semiconductor. Depending on the type of doping, excess electrons or holes are generated in the material which are free to conduct electricity. The impurity atoms frequently employed to dope pure silicon or germanium are the elements of Group III and Group V of the periodic table. These impurity atoms are referred to as acceptor or p-type and donor or n-type impurities as they contribute excess holes and electrons respectively to the semiconducting material. The semiconductor is accordingly known as p-type or n-type semiconductor. The dopants are added in the ratio of about 1 in 10^6 to 10^8 atoms of the semiconducting material. Such a small quantity of dopants does not bring about any structural changes in the semiconductor as the impurity atoms

replace the regular atoms in the crystal. However, the conductivity of the semiconductors is greatly affected by such substitutions.

7.3.1 Donor or n-type Semiconductor

When a pentavalent impurity atom of Group V, such as phosphorus, arsenic or antimony, is introduced into silicon, four of its five valence electrons form covalent bonds with the neighbouring four silicon atoms while the fifth valence electron remains loosely bound to its nucleus as shown in Fig. 7.3a. A small but definite amount of energy is required to detach this fifth electron from its nucleus and make it free to conduct. The energy required is, however, quite small as compared to the energy required for breaking a covalent bond and can be easily provided by thermal agitation inside the crystal.

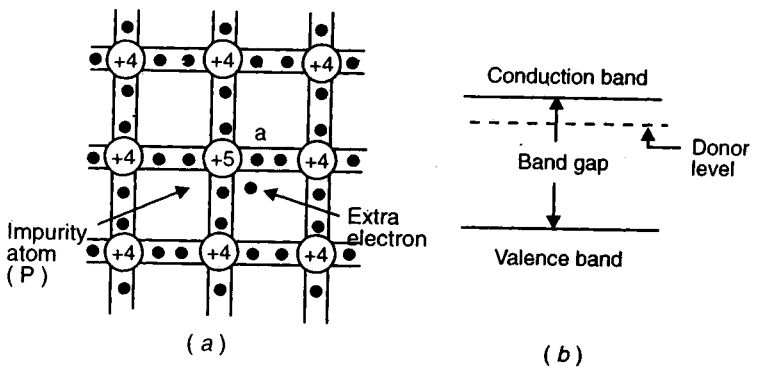


Fig. 7.3.*(a) A pentavalent impurity atom (P) in a silicon crystal. (b) Energy level diagram of an n-type semiconductor.*

The energy level corresponding to the fifth valence electron lies in the band gap just below the conduction band edge as shown in Fig. 7.3b. This level is called the *donor level.* The depth of the donor level below the conduction band is merely about 0.01 eV for Ge and 0.03 eV for Si. The electrons are, therefore, easily transferred to the conduction band leaving behind positively charged immobile impurity ions. Thus each pentavalent impurity atom donates one free electron to the semiconductor. Such impurities are, therefore, known as *donors or n-type impurities* and the semiconductor containing such impurity atoms is known as an *n-type semiconductor.* In these semiconductors, the current is carried mainly by electrons which are called *majority carriers.* The thermally generated holes are called *minority carriers.* The electron concentration, n, is obviously quite large as compared to hole concentration, p, but their product always remains constant.

i.e., $$np = n_i p_i = n^2_i \qquad (7.2)$$

where n_i and p_i are the intrinsic values of the carrier concentration. This relationship is called the *law of mass-action* and will be derived later.

7.3.2 Acceptor or p-type Semiconductor

If a trivalent impurity atom of Group III, such as boron, aluminium, gallium or indium, is introduced into silicon, it forms three covalent bonds with the neighbouring three silicon atoms while the fourth bond is not completed due to the deficiency of one electron. This incomplete bond is shown by broken line in Fig. 7.4a where the small circle (marked 'a') represents the electron deficiency. Thus the trivalent impurity atom has a tendency to accept one electron (say 'b') from a neighbouring silicon atom to complete the fourth covalent bond. This process requires a small amount of energy which is easily provided by the thermal agitation in the crystal. The transferred electron leaves behind a broken covalent bond, i.e., a hole, at position 'b' on the silicon atom which acts as a current carrier.

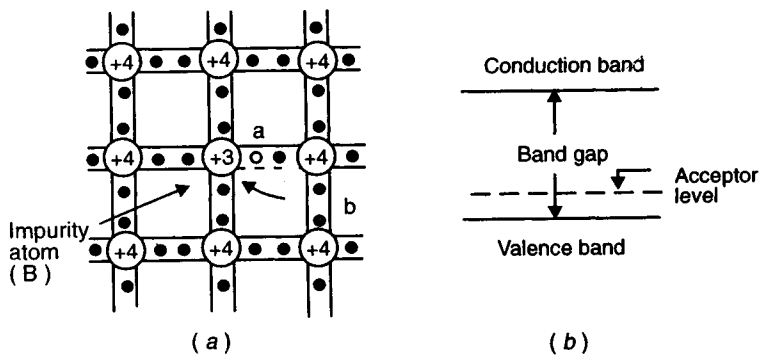


Fig. 7.4. *(a) A trivalent impurity atom (B) in a silicon crystal. (b) Energy level diagram of a p-type semiconductor.*

The energy level corresponding to the electron deficiency of the type 'a' is located just above the valence band and is called the *acceptor level.* The acceptor levels are located at a distance of about 0.01 eV above the top of the valence band in Ge and at about 0.046 to 0.16 eV in Si. An electron can be easily transferred from the valence band to the acceptor level by providing this small amount of energy. This creates a hole in the valence band which acts as a mobile current carrier. The negatively charged impurity atom, however, remains immobile and does not contribute to conduction. Thus each trivalent impurity atom can accept an electron from a neighbouring silicon atom to produce a hole in the semiconductor. Such impurities are, therefore, known as *acceptors or p-type impurities* and the semiconductor containing such impurity atoms is known as a *p-type semiconductor.* In these semicon-